Sunton ESP32-8048S043C connectors (1.25 mm, 1:1 cables).
Check the cable maps pin 1 to pin 1 before power-up.

3.3 V for the scanner: the SparkFun header takes 3.3 V, not 5 V.

SparkFun 2D Barcode Scanner Breakout (DE2120, red aiming line).
Plugs in on top; M3 x 11 mm standoff in H1.
GM65 instead: J4. Fit only one scanner.

PWR_FLAG
+5V
PWR_FLAG
GND

J1
GND 1
SCAN_TX 2
SCAN_RX 3
+5V 4
Sunton P1 UART

AMS1117-3.3
U1
+5V 3
VI
VO 2
+3V3
GND 1

C4 22uF GND +5V
C1 10uF GND +5V
C2 22uF GND +3V3
C3 100nF GND +3V3

J3
SCAN_STAT 1
SCAN_TRIG 2
+3V3 3
SCAN_TX 4
SCAN_RX 5
GND 6
SparkFun 2D Barcode Scanner

TP1 TRIG SCAN_TRIG
TP2 STAT SCAN_STAT

J2
BUZZ 1
LED_3V3 2
3
GND 4
Sunton P4 IO

Status LED: WS2812B, data shifted 3.3 V -> 5 V (IO17).

U2 74AHCT1G125
VCC 5 +5V
GND 1 GND
LED_3V3 2
4 LED_5V
GND 3 GND
C5 100nF GND +5V

R3 330R LED_5V LED_DIN

D2 WS2812B
LED_DIN 4
DIN DOUT 2 LED_DOUT
VDD 1 +5V
VSS GND
C6 100nF GND +5V

TP3 LED_DOUT
LED_DOUT

Buzzer: MLT-8530 from 5 V, PWM on IO18 (ESPHome rtttl).

LS1 MLT-8530
+5V 1
BUZZ_SW 2

D1 1N4148W
+5V
BUZZ_SW

R1 100R BUZZ BUZZ_G
R2 100R BUZZ_G GND

Q1 IRLML6400A
BUZZ_G 1
BUZZ_SW 3
GND 2

J4
GND 1
+5V 2
SCAN_TX 3
SCAN_RX 4
GM65 scanner

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H1 M3 standoff (scanner)
H3 M3 (case)
H4 M3 (case)

Sheet: /
File: scanner-carrier.kicad_sch

Title: Pantry scanner carrier

Size: A4	Date: 2026-10-27	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1